Tubesense: GPS-free train positioning from phone rides, with a verified agent layer

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Abstract— Parts of the London Underground are loud, but TfL does not publish section-level noise or grinding data. We record ordinary rides on one phone and place the train between stations without GPS, by matching the gyroscope's rotation about gravity against maps built from earlier rides. In a test written down before the rides (pre-registered), on new Central line rides with frozen maps, the median position error was 1.95 m against 8.2 m for a timetable estimate, with 96 % coverage of the 90 % band. A Gemini 4 Argon agent layer tunes the tracker against a scorer it cannot edit, reads TfL and London Assembly records stored in MongoDB Atlas, and drafts a cited evidence pack in which a deterministic verifier checks every quote and number; a human approves the pack before use. We report the negative results as plainly as the positive ones: phone audio could not reliably rank the two sections by loudness, our corrugation method failed, and the tracker did not generalise to a second line on the data we had.

1. Problem and user

TfL's own in-cab measurement puts Liverpool Street→Bethnal Green (Central line, eastbound) at 87 dB LAeq and Bethnal Green→Mile End at 83 dB (24 June 2024, FOI-1231-2425); a press report gives 92.8 dB, whose primary source we could not locate. Both sections were ground for noise between December 2021 and May 2022 (Mayor's Question 2022/1791), and TfL says grinding must be repeated "usually 6 months – a year" (TfL Tube Noise presentation, 6 December 2022). Section-level data is not public, and at Mayor's Question Time on 16 July 2026 (2026/2156) Elly Baker AM said TfL had turned down most of her data requests. Our user is her office: it needs measurements that can be placed on the track and a pack of claims it can check line by line.

2. Data

- **Rides:** 9 recordings on 23 October 2026 (Victoria and Central lines) and, on 24 October, 6 Central and 3 Hammersmith & City rides, 2 NIOSH-app trips and a street-level magnetometer calibration. iPhone 12, phyphox 1.2.1 (accelerometer and gyroscope at about 99 Hz, magnetometer, barometer at about 1 Hz), Voice Memos audio.
- **Messy inputs, detected from the data:** a phone on the floor (tilt and vibration), two phyphox pauses (recovered from its event log), a trip split across two files, calibrated versus raw magnetometer runs, iOS recalibrating the magnetometer mid-ride, and sync taps that were silent in the audio (sync falls back to matching vibration against sound).
- **Privacy:** audio is decoded in memory and never written to disk; only 1-second A-weighted levels and third-octave bands are kept, with random trip ids. Gemini receives only public documents and summary statistics.
- **Public records:** 14 TfL and London Assembly documents (44 chunks) plus TfL working-timetable section lengths.

3. Method

3.1 Positioning

The gyroscope's yaw rate about the gravity vector is the most position-locked signal: curves turn the phone the same way on every ride. Offline, a ride is aligned to a map of earlier rides with dynamic time warping (DTW) in the distance domain, with speed forced to zero at both stops. The live tracker is a causal hidden Markov model over position along the section that uses only data up to the current second and reports a 90 % band. Maps are built leave-one-ride-out, so a ride is never tracked against itself.

3.2 Evaluation

There is no GPS truth in a tunnel. Truth comes from physical landmarks: the start and end of each curve, matched across rides, and scored by interpolation between landmarks without DTW, so the reference does not share the tracker's method. We also hide the gyro around each landmark to measure positioning between curves. The 24 October analysis (hypotheses H1–H5 and their pass/fail rules) was written down and frozen at git tag `prereg-24oct` before those rides; anything decided later is labelled post-hoc.

3.3 Noise levels

1-second A-weighted levels are relative to the phone's digital full scale (uncalibrated) and are compared only within the same trip. Two indicative absolute readings come from the NIOSH Sound Level Meter app (validated only up to about 85 dB(A)).

4. Results

Table 1: Results. The 24 Oct rows were tested under rules fixed before the rides.

Question	Verdict	Evidence
Gyro curve signatures repeat ride to ride (Central, 23 Oct)	**GO**	same-section r 0.76–0.99 vs 0.20 for different sections
Offline positioning (DTW on yaw)	**GO**	held-out curve edges: median 6 m vs 24 m for a time-fraction baseline
Live tracker, held-out rides (23 Oct)	**GO**	median 2.2 m vs 11.2 m timetable; 90 % band coverage 0.92. Between curves no better than the timetable (11 vs 16 m)
H2b, pre-registered: frozen maps on new Central rides (24 Oct)	**GO**	Liverpool St⇄Bethnal Green median 1.95 m vs 8.2 m timetable, coverage 0.96, 6 sections (rule: ≤ 5 m and ≥ 0.80)
H1/H2a: second line, Hammersmith & City	***NO-GO***	one usable pair (r 0.72 vs null max 0.84); tracker 52 m vs timetable 47 m
Section loudness ranking from phone audio	***NO-GO***	Liverpool St→Bethnal Green louder on 10 of 11 trips, mean +0.54 dB; the frozen rule needed every trip (R10 westbound reversed, −0.13 dB); the difference is also inside the recorder's limiter compression
Magnetometer as a position fingerprint	***NO-GO***	|B| spoiled by recalibration and bias; H4: same-platform sd 19 vs between-station 22 µT, no stable platform signature
H5: absolute level, NIOSH app	**INDICATIVE**	LAeq 91.9 dB(A) eastbound, 87.3 dB(A) westbound, one whole trip each; may read low above 85 dB(A)
Corrugation wavelength from the roar	***NO-GO (method)***	dominant peaks stay at fixed frequencies as speed changes (9 of 9 runs), so frequency ÷ speed gives no wavelength; corrugation as the cause is untested by us
Tracker v2 (post-hoc, tuned by the Argon analyst)	matched, did not beat	single sealed score on 24 Oct rides: median 1.8 m, coverage 0.936, vs v1 1.8 m / 0.951; its stop gate never engaged on real stops

5. Agent layer (Gemini 4 Argon, thinking level medium)

- **Analyst.** Proposes allow-listed tracker settings, each with a hypothesis that can be proven wrong; a harness it cannot edit scores them on the 23 October rides and a rule fixed beforehand accepts or rejects (6 experiments, 1 accepted). Every call is written to an append-only hash-chained log mirrored to Atlas; the harness allows exactly one scoring on the 24 October rides (result in section 4).
- **Investigator.** Reads the whole corpus (long context) plus Atlas hybrid-search hits and our statistics, and returns typed claims (sourced, measured, derived, inference). A deterministic verifier requires every quote to match its source verbatim and every number to trace to a quote, a statistic or a recomputed formula, and rejects known overclaims. On 25 October: 15 claims, 5 rejected and fixed. A review against the full sources then found 21 framing issues the verifier cannot see (4 major); after revision, 16 claims, all verified, none dropped, plus 4 information requests.
- **Ask Argon (live).** The same verifier loop on 2–5 claims, then an Argon review of the claims against their cited sources; the one-line answer is checked in code for unsupported numbers and overclaims (for example a ranking from our NO-GO data).
- **Roughness reader.** Reads the chart images in TfL's 2018 rail-roughness FOI PDF (Bethnal Green→Mile End). Its readings matched a model-free pixel reader on 120 of 120 values; both runs find a rough stretch at about 725–905 m, overlapping TfL's own statement of elevated roughness between 720 and 880 m. The checks caught misnamed line colours, readings that did not match the pixels and numbers Argon calculated itself; those were dropped. Pending a human check by eye.
- **Recheck.** With code execution, Argon re-derives our headline statistics from summary tables and the frozen rule text, never seeing our answers: 65 of 65 values match within tolerances fixed in advance.
- **Why Argon.** Over 6 runs per model, gemini-3.8-flash did as well on the verifier loop (same rejection rates, time and final counts). Argon used more of the record (for example the Mayor's transparency promise in 5 of 6 runs vs 0 of 6) but made more framing errors in long cross-document claims (one reviewer, not blind). Hence the human approval step.

6. MongoDB Atlas and human approval

An M10 cluster (MongoDB 8.0) holds the corpus with Atlas Search (full text), Vector Search with Automated Embedding (Voyage `voyage-4`), and Hybrid Search (`$rankFusion`), which ranks the expected document first

on 12 of 12 test questions (written by us, not blind). Atlas Stream Processing reads rides arriving one row per second in `ingest_live` (a tracker replay, or live phone sensors read over phyphox remote access) and writes 10-second windows to `live_view`. Atlas also stores the hash-chained logs, claims, asks and signed approvals. Approval needs a five-item checklist and a name, and re-runs the verifier at that moment; it creates a SHA-256-signed draft and an Atlas record. Nothing is sent automatically.

7. Limitations

- One phone, two mornings, 2–3 rides per section; Central line positioning only.
- Phone audio levels are relative and compressed by the recorder's limiter; the NIOSH readings are indicative.
- Position truth is a landmark reference, not GPS; between curves the tracker is no better than a timetable.
- The roughness readings describe TfL's 2018 data on one section, before the 2021–22 grinding, and await a human check.
- Framing that the verifier cannot check is caught only by review and the human approver.

8. Next steps

More rides, lines and phones; a calibrated reference meter; comparing positioned noise with TfL's roughness plots; the React Native capture app (built and tested off-device, not yet run on a phone); and a purpose-built instrument concept (Research Box v2, not built) in which each part targets a limit we measured.

Prior work. Royal Holloway 2019, Sensors 19(19):4184 (accelerometer DTW, station-level); Royal Holloway 2017, arXiv 1704.00148 (magnetic distortion underground); MLoc, arXiv 2003.10531; Dieckow et al. 2025, arXiv 2507.19327 (magnetic map-matching with under-train sensors); Huyan et al. 2023, JMIR, PMC10686533 (smartphone sound level meter validity). All numbers in this paper are in `outputs/FINDINGS.md`, `README.md` and `outputs_posthoc/` of the repository.